Roll Number: _____

Thapar Institute of Engineering & Technology, Patiala, Punjab

Department of Computer Science and Engineering

B.E. MST Examination (Summer Semester), June 2026

Course Code: UCS520 & UCS414	Course Name: Computer Networks
Date: 28th June, 2026	Time: 3:00-5:00 (Evening)
Duration: 2 Hours, Max. Marks: 30	Name of Faculty: Dr. Himanshu Gauttam

Note: Attempt all questions. Attempt all parts of one question at one place. Assume missing data if any.

<table>
<tr><th>Q. No.</th><th>Questions</th><th>Marks</th><th>CLO</th></tr>
<tr><td>Q.1</td><td>(A) Describe the key components of a communication network with the help of a suitable diagram?

(B) An organization is assigned with the following IP address 200.1.2.0/24, which is further divided into 4 equal subnets. Based on the given information, determine the followings:
(i) Subnet mask for the third subnet
(ii) IP address of the second host of second subnet
(iii) IP address of the last host of first subnet
(iv) Direct Broadcasting Address (DBA) for the second subnet
(v) IP address of the second host of third subnet
(vi) Range of the usable hosts in fourth subnet

(C) Describe the services provided by the Transport, Network, and Presentation layers of the OSI model.</td><td>(4)

(3)

(3)</td><td>L2

L3

L1</td></tr>
<tr><td>Q.2</td><td>(A) Explain the hierarchy of interconnecting devices a suitable diagram. Further, list out the key differences between the layer-2 switch and router.

(B)A communication network with 1 MBps bandwidth, is shared among four stations using bus topology. The stations utilize the ALOHA protocol as media access protocol. Each station transmits the data frame of size 1000 KB. The data transmission instance of the stations is listed in table:

<table><tr><td>Station ID</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Data transmission Instance</td><td>0.5 Sec</td><td>2.2 Sec</td><td>3.1 Sec</td><td>4.2 Sec</td></tr></table>
It is assumed that there is no other active station present the in the communication network except the above listed four stations. Based on the above listed information, determine that how many stations will attain the successful data transmission without a collision.</td><td>(4)

(6)</td><td>L2

L4</td></tr>
<tr><td>Q.3</td><td>(A) For the given digital data sequence: 0 0 1 0 1 1 0 1 1 0, draw the digital signals using NRZ-I, Manchester, Differential Manchester and Pseudo ternary line coding schemes.

(B) What are the different persistence schemes used in CSMA. Discuss the working of each scheme with suitable state-space diagram. Discuss the key limitations of CSMA/CD protocol in the context of wireless networks.

(C) Explain the major drawbacks of the Stop-and-Wait-ARQ protocol.</td><td>(4)

(5)

(1)</td><td>L3

L2

L2</td></tr>
</table>